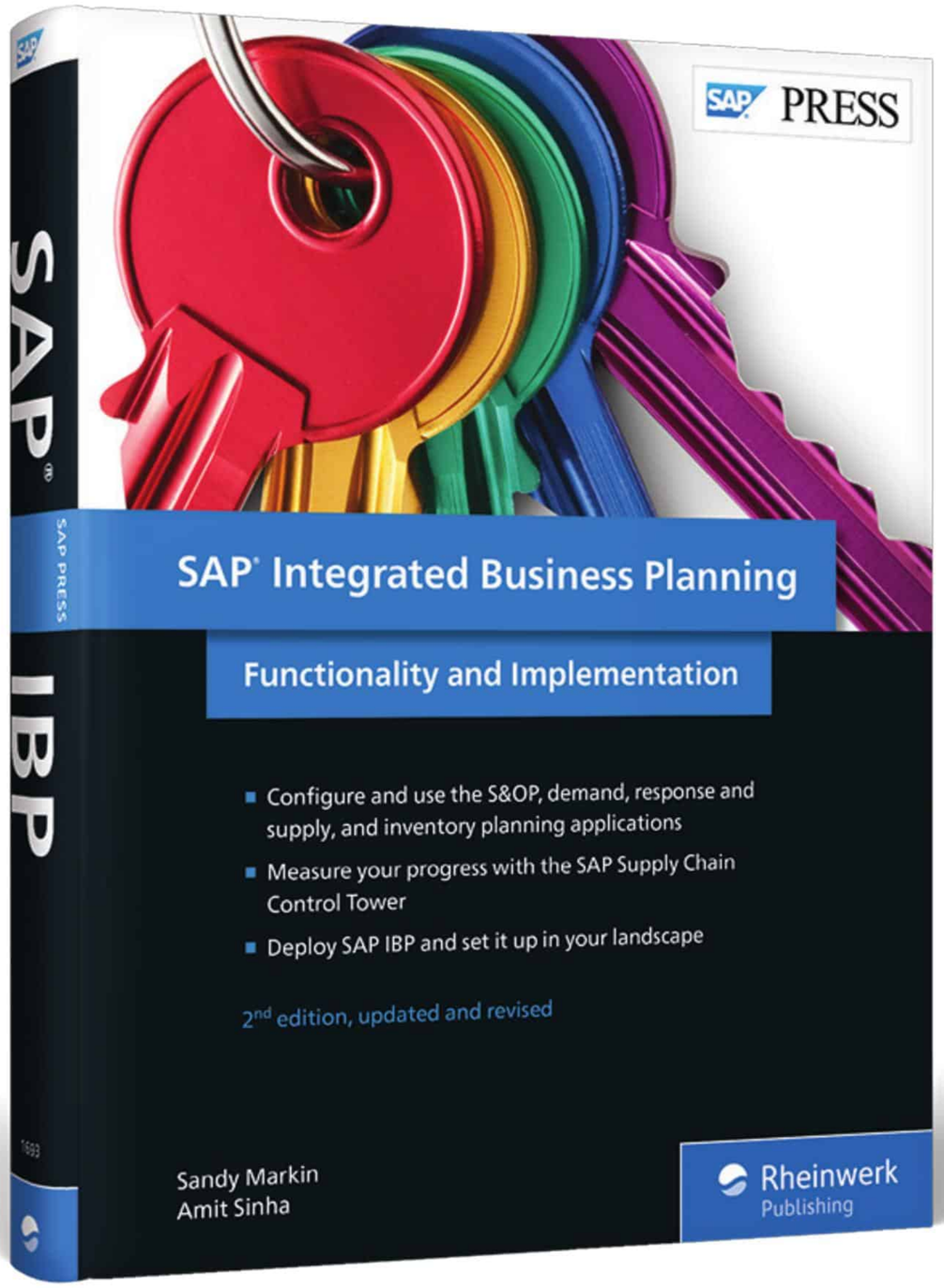
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Reading Sample

This sample chapter discusses the concepts behind supply planning and response management in SAP Integrated Business Planning for response and supply. It also covers rule-based response and supply planning with demand prioritization, supply confirmation, and allocation.

"Response and Supply Planning with SAP IBP"

Contents

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The Authors

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Chapter 10

Response and Supply Planning with SAP IBP

Customer demand, finalized through the demand planning process, is used for driving supply through production, transportation, and purchasing. This chapter discusses the concepts behind supply planning and response management business processes along with their modeling in SAP Integrated Business Planning (SAP IBP) for response and supply.

The process of planning for fulfillment in the supply chain is called supply planning and is performed to meet requirements such as anticipated customer demand, sales orders, or internal consumption. An organization fulfills customer demand through a combination of producing, purchasing, and transporting goods and services across the supply chain. This chapter will discuss the various supply planning approaches used to successfully execute these activities.

Supply planning is relevant for short-term, medium-term, and long-term horizons with special focus on the medium-term planning horizon. Long-term supply planning is aligned with sales and operations planning (S&OP), while the short-term supply planning works with response planning.

Response planning, on the other hand, focuses on reacting to market changes in the short term for the most desirable results. When the actual event varies from the anticipated behavior, a response management solution is required. The real demand of customers may vary from the forecast; a supplier's inability to supply a component or a transportation delay can disrupt the original supply plan. An effective response management solution enables the organization to determine the best course of action in those scenarios. Therefore, response planning is a supply solution with a short-term horizon.

In this chapter, we'll start with an overview of response and supply planning, followed by a discussion of the response and supply planning methodology. Building

on the knowledge gained on supply planning in Chapter 6, Section 6.3.2, and Chapter 7, Section 7.3.2, we'll move into a discussion of rule-based response and supply planning with demand prioritization, supply confirmation, and allocation. SAP IBP for response and supply includes a highly valuable feature: gating factor analysis, which we'll discuss in a separate section. Finally, we'll analyze the planning view, simulation, and collaboration applications in SAP IBP for response and supply.

10.1 Response and Supply Planning Overview

Table 10.1 shows the supply planning solution in short-term, medium-term, and long-term horizons. Long-term supply planning is normally performed for 18 months to 5 years and is typically in a monthly horizon. This perspective drives the strategic decisions of the organization for demand and supply. Decisions regarding capacity enhancement, product development, supplier selection, and network changes are examples of long-term supply planning. Mid-term supply planning decisions are operational in nature and generally performed in weekly buckets for a typical period of 6 weeks to 18 months. Operational supply planning of production, distribution, and procurement are performed for this period. Short-term supply planning or response planning is generally performed in daily buckets for a period up to 6 weeks.

Planning Horizon	Short Term	Medium Term	Long Term
Representative time bucket	Daily	Weekly	Monthly
Strategy level	Response	Operational	Strategic
Planning activity examples	■ Adjust supply plan ■ Respond to incoming customer orders for deviation from forecast ■ Confirm sales orders ■ Plan allocation for constrained supply	■ Develop production, purchasing, and stock transfer plans against demand forecast and sales orders ■ Forecast-driven replenishment plan ■ Action for resolution of any predicted supply issue	■ Mid- to long-term higher-level supply plan ■ Supplier collaboration ■ Executive and team collaboration to have one organization-level plan

Table 10.1 Supply Plan for Different Planning Horizons

The start of a response planning period can be based on any frozen period. For organizations that follow a frozen period ranging from a couple of days to a week, the response solution is relevant from the end of the frozen period to the end of the response solution horizon. For a frozen period of 1 week and response solution period of 6 weeks, the horizon of response planning would be from 1 to 6 weeks. With no frozen period, the response solution horizon would be from the current date to the end of the response planning horizon. Typical decisions for the response planning solution are supply plan adjustments, order fulfillment, and quota assignment.

For most industries, demand is volatile, lead time for product availability is higher than the required customer service time, and carrying inventory increases costs and reduces organizational flexibility. These factors demonstrate the need for efficient response and supply planning. For organizations with a presence across the globe and a complex network structure of production plants, warehouses, suppliers, and customers, efficient response and supply planning is paramount for efficient customer service. An efficient response and supply solution enhances the profitability of the organization by adding value through the following:

- Higher service rates
- Lower inventory costs
- Alignment with business priorities
- Higher resource utilization

Lead times for procuring and producing a material is often higher than the time anticipated by the customer to receive the material after placing an order. Thus, supply planning in advance is crucial for an organization to be able to serve a customer with a higher service rate in both *make-to-stock* and *make-to-order* environments. In make-to-stock scenarios, ideally, the material should be available in finished goods inventory before receiving the customer order; in make-to-order scenarios, the required components should be available in the finished goods inventory to start the final assembly or manufacturing as soon as the customer order is received. At the same time, carrying inventory locks capital, increases costs, and reduces organizational flexibility. Therefore, serving the customers optimally and responding to changing market realities against forecast becomes hugely important, and a response and supply solution becomes the central nervous system of the organization.

Not all demands are equal for an organization, and you may be required to follow a prioritization rule when fulfilling orders. An efficient response and supply solution

must meet this requirement while aligning demand and supply because these capabilities increase an organization's service level and profitability in the long run.

10.2 Supply Planning Methodology

The supply planning algorithm creates supply plans for production, purchase, and stock transfer, driven by the master data of the supply chain network and the algorithm type selected for the supply plan. The master data that governs the supply planning solution are the production rule (driven by bills of materials [BOMs], resource, and recipes/routing); the replenishment rule (driven by the sourcing logic or transportation lane); and the network structure for products (through production rule, transportation rule, or procurement rule). Master data for supply planning is discussed in detail in Chapter 3.

The following time series-based planning algorithms are available:

- **Time series-based supply planning heuristic**
  You can use this algorithm for infinite demand and supply planning. The algorithm calculates supply based on the assumption that all resource capacities are infinite. The plan will fulfill all demands, balancing insufficient supply with negative projected stock for the product/location combination in question. This algorithm does not necessarily produce a feasible plan, but you can use the recommended plan to identify issues with supply and capacity bottlenecks.
- **Time series-based supply propagation heuristic**
  As a variant of the time series-based supply planning heuristic, this algorithm also calculates an infinite demand and supply plan. In contrast, however, this algorithm only propagates the available supply downstream through the supply chain and does not fulfill all demands. Therefore, the plan will allow you to see the impact of supply problems, that is, when there will be customer demands and net demands for specific customer/product and location/product combinations that will not be met.
- **Time series-based supply planning optimizer**
  You can use this algorithm to generate a cost-optimized production, distribution, and procurement plan for the entire supply chain network, while taking into account certain constraints. The algorithm can be configured to generate a plan that maximizes either profits or delivery.

The following order series-based planning algorithms are available:

- Supply, allocation, and response planning
- Order confirmation planning
- Deployment planning

Supply, allocation, and response planning are the backbone of SAP IBP for response and supply. Constrained supply, allocation, response, and order confirmation planning are performed through two different planning operators:

- Supply and allocation planning
- Response and order confirmation planning

For supply, allocation, and response planning, you have the option of fulfilling the demand through a prioritization sequence to meet higher-priority demands during times of constrained supply first. The next section is devoted to the demand prioritization logic in SAP IBP for response and supply before going into the details of the planning logic we'll describe.

Unconstrained heuristic supply planning and constrained supply optimization are the supply planning algorithms for time series-based planning. These algorithms are part of the planning operators of SAP IBP for sales and operations and are covered in detail in Chapter 7. Optimization in order series-based planning is currently on the product roadmap of SAP IBP for response and supply and should be available in 2018. However, its functionally will be similar to the optimization concepts described in Chapter 7.

10.2.1 Demand Prioritization

Demand prioritization in SAP IBP is achieved through a prioritization rule, segments for the rule, and sorting logic within a segment. The elements used to calculate demand prioritization are as follows:

- Demand prioritization rule
- Demand segments
- Sorting condition in a segment
- Review of demand sequence on defined priority

Figure 10.1 shows an example of the demand prioritization logic in SAP IBP through prioritization rule, segment creation, and sorting criteria. The prioritization rule

prioritizes every demand element in a planning version or a subset of the elements based on certain selection criteria. Each rule must have at least one segment. Segments are the building blocks of the rule; a rule can have multiple segments, and the sequence of the segments defined by their prioritization. A segment defines the selection criteria or condition for a demand element. Based on the selection criteria and property of a demand element, the demand element is associated with a segment.

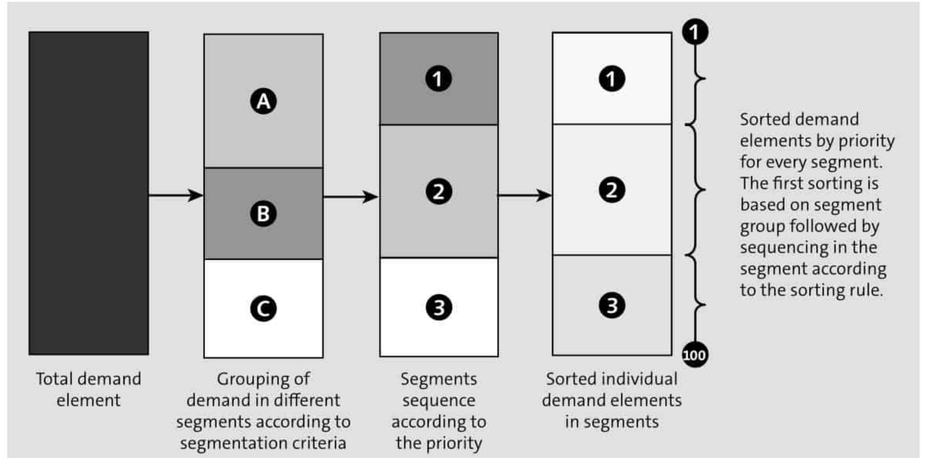


Figure 10.1 Demand Prioritization through Segments and Sorting Criteria in the Prioritization Rule

For example, a segment can be created for a customer classification (e.g., customers classified as Ⓐ, Ⓑ, or Ⓒ); material categorization (e.g., materials categorized as group ❶, ❷, ❸, etc.); material requirements planning (MRP) for demand type (e.g., MRP type VC for sales orders, FC/FA for forecasts, etc.); or any other segments based on the attributes defined for the master data elements. The planning system identifies the demand elements as part of the segment group and then processes the segments per the sequence identified in the rule. Each segment can have one or multiple sorting criteria for further sequencing of the demand elements within a segment. Thus, within a segment for customer group A, individual demand elements can be sorted based on attributes such as requirements date, order delivery priority, revenue, order entry date, etc. Sorting grouped by segment identifies and ranks every demand element in the system in order of replenishment priority for the response and supply planning solution. Planners can review demand elements, with their prioritization

ranks, are available for review in the SAP IBP system. Based on the review, the prioritization rule can be edited to meet business priorities.

10.2.2 Supply, Allocation, and Response Planning

Rule-based response and supply planning in a constrained environment is achieved through supply allocation and response planning algorithms in SAP IBP for response and supply. Figure 10.2 shows the planning sequence and details for rule-based supply, allocation, and response planning. Prioritized demand through the prioritization rule and supply constraints (resource, material, supplier capacity, etc.) are the inputs for *constrained supply planning*. A constrained supply planning run is also known as a *constrained forecast run* in SAP IBP. The output of this planning run is a supply plan and a *constrained demand forecast*. A constrained demand forecast is generated using the prioritized demand and organization's ability to supply while considering the supply constraints. In a short-supply scenario, the constrained demand forecast will be less than the unconstrained forecast, as shown in Figure 10.2.

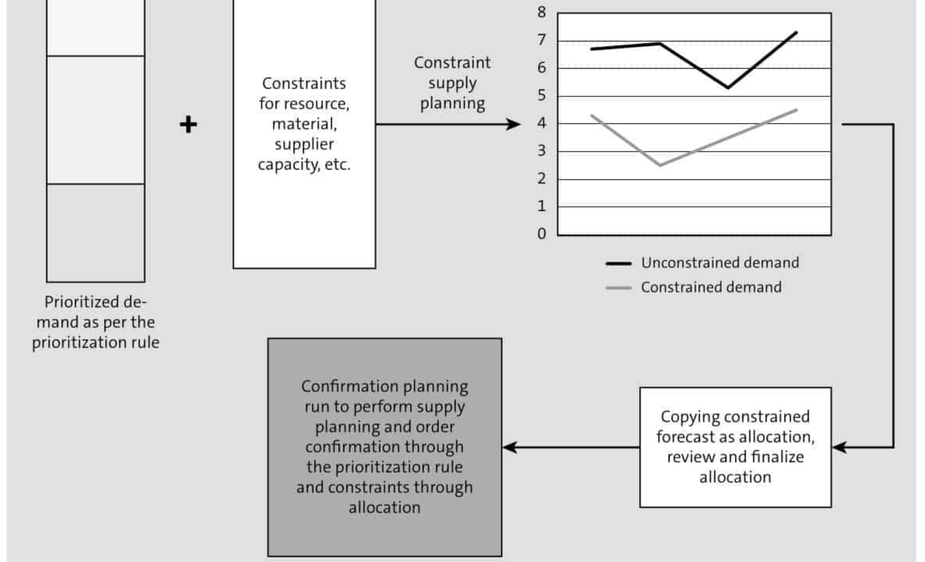


Figure 10.2 Supply Allocation and Response Planning with SAP IBP for Response and Supply

In SAP IBP, the constrained forecast can be copied as the product allocation, which will limit the supply of a product against an order while considering the prioritized demand. A confirmation planning run, available as a separate algorithm in response and supply planning, re-plans the supply with the most recent prioritized demand and the product allocation. This planning run also performs the order confirmation process. For any change in the demand and supply situation, the confirmation planning run re-creates the plan and confirms the orders. Therefore, the output of this complete cycle are the confirmed orders and the most preferred supply plan based on the prioritization rule and organization's ability to supply.

10.2.3 Deployment Planning

Deployment planning is performed as short-term supply planning to distribute the available supply to demand. Deployment is normally performed only in the short term for the *actual product movement horizon*, typically from a couple of days to a week. Deployment planning can be grouped with a planning time fence or frozen horizon for supply planning. The deployment run takes all customer demand (sales orders); all confirmed component demands (for example, from production orders or stock transfer orders); all forecasted customer demands; and safety stock into account. The system does not look at the possibility of creating planned orders or purchase requisitions but tries to satisfy demand with existing supply elements. Stock transfer requisitions that are pegged to supply elements and are considered available to deploy become deployment stock transfer requisitions. After a deployment run has been executed, deployment stock transfer requisitions are created, which consume confirmed supply elements. These deployment stock transfer requisitions can be transferred to SAP ERP or SAP S/4HANA and used in execution processes to prepare the physical transport of goods.

10.3 Forecast Consumption

Forecast consumption is a standard requirement and functionality of response and supply planning. In make-to-stock scenarios, a product is manufactured and then made available in inventory in anticipation of a customer order. After a sales order is entered, the SAP IBP system consumes an existing forecast first. This consumption is usually performed at either the period/product/location or the period/product/location/customer level. You have the option of using planning levels for the flexibility to either consider or not consider the customer while consuming a forecast.

Table 10.2 shows an example of forecast consumption with the logic of the consuming forecast in the same bucket against the requested sales order. The total demand on a material in a time bucket is the sum of all requested sales orders and the net forecast (unconsumed forecast) in the time bucket. Thus, if a requested sales order in a time bucket is less than the forecast, the total demand in that period is the original forecast. However, if the sales order is higher than the forecast in a time period, then the net demand is equal to the total number of requested sales orders.

Time Period (Months)	1	2	3	4
Unconstrained forecast	100	100	100	100
Requested sales order	60	120	90	110
Consumed forecast	60	100	90	100
Net forecast (unconsumed)	40	0	10	0
Total demand (sales order + forecast)	100	120	100	110

Table 10.2 Forecast Consumption with Consumption in the Same Time Bucket Only

10.4 Gating Factor Analysis

Gating factor analysis provides information on all the constraints that may prevent the fulfillment of a sales order on time. Gating factors are associated with the order details in response and supply planning. An order may not be fulfilled on time due to limitations or constraints such as resource availability, material shortage, allocation, etc. The gating factor functionality in SAP IBP provides detailed information on the limiting factors so that the planner can analyze the issue in advance and take appropriate action.

Gating factor types and related information in SAP IBP for response and supply are shown in Table 10.3. This ready reference information in the system helps you perform root-cause analysis on unwanted scenarios before the exception actually hits the demand and supply plan. This alert mechanism is highly useful in adjusting the plan to take the best possible action in the event of supply challenges.

Gating Factor Type	Gating Factor Information
Lead time	Procurement or stock transfer lead time
Resource	Production capacity availability of a resource
Projected stock	Stock level lower than the expected level to fulfill an order based on the demand and supply situation
Supplier constraint	Supplier commitment with the limit on the material supply in a horizon
Product allocation	Allocation finalized by the business in a supply shortage scenario
Supply chain model	Issue in supply chain model consistency and data elements

Table 10.3 Gating Factor Types in SAP IBP

10.5 Order Review and Analysis

Order review and analysis in SAP IBP for response and supply is performed at both the time-bucket aggregate level as well as at the detailed order-based level. Planning results in time-bucket aggregation can be reviewed in the SAP IBP Excel planning view. Order-level reviews for the order confirmation, projected inventory, and gating factor analysis are performed in the SAP Fiori launchpad for SAP IBP.

10.5.1 SAP IBP Excel Planning View

Even though the response and supply solution is based on order-level planning, planning key figures of demand, supply, allocation, and net calculations can be reviewed in the time-bucket planning view. Based on its properties, a key figure may be available for review but not for editing. Key figure information normally relevant for response and supply planning in the Excel planning view includes demand forecast, allocation, constrained forecast, inventory, net demand, production, purchase, and projected inventory.

Figure 10.3 shows an example of the SAP IBP Excel planning view relevant for response and supply planning. The forecast, sales order, and allocation planning view example shows these key figure in the daily time bucket. Based on business requirements and the nature of the organization's business, a weekly or monthly time

bucket can also be considered for this review. The second example in the same figure shows the projected stock view, along with elements of total demand, total supply, and inventory information. Note that the key figures not relevant for change in the planning view are grayed out with a different background color than white.

Forecast, sales order, and allocation planning view example

Customer ID	Location ID	Material Number	Key Figure	12/05/2016	12/06/2016	12/07/2016	12/08/2016	12/09/2016	12/10/2016	12/11/2016	12/12/2016
10000012	2000	PR1_ABC	Allocation	100	100	50	40	100	100	100	100
			Allocation Adjusted			**60**					
			Allocation Consumed	50	100	50	40	0	0	0	0
			Allocation Final	100	100	60	40	100	100	100	100
			Allocation Remaining	0	0	10	0	100	100	100	100
			Forecast Constrained	100	100	50	40	100	100	100	100
			Forecast Unconstrained	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
			Sales Order (Confirmed)	50	100	50	40	0	0	0	0

Projected stock planning view example

Location ID	Material Number	Key Figure	11/16/2016	11/17/2016	11/18/2016	11/19/2016	11/20/2016
DC1	PR_AC2	Stock Projected	12	62	82	-18	0
		Initial Stock	0	0	0	0	0
		Total Demand	50	100	50	100	100
		Total Receipt	62	150	70	0	118
		Production (Planned)	0	0	0	0	0
		Production (Confirmed)	0	0	0	0	0
		Distribution Receipt (Planned)	0	50	70	0	118
		Distribution Demand (Planned)	50	100	50	100	100
		Distribution Receipt (Confirmed)	62	100	0	0	0
		Distribution Demand (Confirmed)	0	0	0	0	0
		Dependent Demand (Planned)	0	0	0	0	0
		Dependent Demand (Confirmed)	0	0	0	0	0
		Safety Stock	0	0	0	0	0

Figure 10.3 Forecast, Sales Order, Allocation, and Projected Stock Planning in the SAP IBP Excel Planning View

Figure 10.4 shows another example of a supply planning view in the SAP IBP Excel planning view for SAP IBP for response and supply: the capacity planning view, which shows available capacity, consumption, and consumption percentage. The supplier planning view shown in Figure 10.4 provides an example of a supplier's delivery plan and its constraints.

Note that the planning view example discussed in this section is just for reference; based on the planning example in SAP IBP for response and supply, any combination of key figures can be added as part of the planning view for review and analysis purposes. Due to SAP IBP's inherent capability showing order series data in the appropriate time bucket, the planning data review process through the Excel planning view can become highly useful for planners.

Capacity planning view example

Location ID	Resource	Key Figure	11/15/2016	11/16/2016	11/17/2016	11/18/2016	11/19/2016	11/20/2016	11/21/2016	11/22/2016
PL1	MC1_PL1	Capacity Available	480	480	480	480	0	0	480	480
		Capacity Available (Adjusted)			400	320				
		Capacity Available (Final)	480	480	400	320	0	0	480	480
		Capacity Consumption	350	350	400	320	0	0	600	700
		Capacity Consumption (%)	73	73	100	100			125	146
		Capacity Consumption (Confirmed)	0	0	0	0	0	0	0	0
		Capacity Consumption (Planned)	350	350	400	320	0	0	600	700

Supplier plan and constraint example

Material Number	Supplier ID	Key Figure	11/17/2016	11/18/2016	11/19/2016	11/20/2016	11/21/2016	11/22/2016	11/23/2016
PR1_AGH	SUPPLIER1	Supplier Constraint	75	75	75	75	75	75	75
		Supplier Delivery (Confirmed)	60	55	50	0	0	0	0
		Supplier Delivery (Planned)	60	55	50	40	75	100	70
	SUPPLIER2	Supplier Constraint	120	120	120	120	150	150	150
		Supplier Delivery (Confirmed)	120	100	100	80	0	0	0
		Supplier Delivery (Planned)	120	110	100	80	150	200	140

Figure 10.4 Capacity and Supplier Plan in the SAP IBP Excel Planning View

10.5.2 SAP IBP: SAP Fiori View

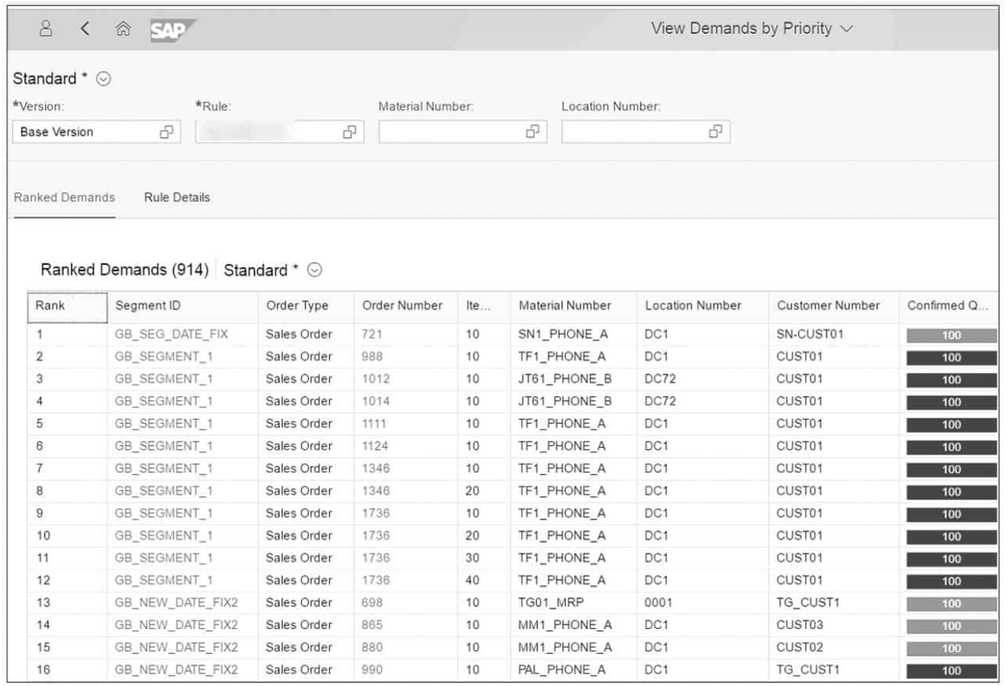
Order-relevant planning information can also be reviewed through the SAP Fiori launchpad for SAP IBP for response and supply. The Demand by Priority app provides a detailed, sequential view of demand per prioritization rule. The View Projected Stock app is used to analyze the projected stock of all materials in the supply chain. This analysis can help planners select products with shortage potential as well as identify extra inventory in the supply chain, which increases cost and reduces flexibility. This application has options to review the demand and supply situation in an element view, a periodic view, or a chart format. Order confirmation information can be analyzed through the View Confirmation app in SAP IBP. For material, customer, and order information, details are available for the percentage of the order confirmed requested date, quantity, and delay in order fulfillments with date and quantity. Gating factor analysis, on the other hand, provides root-cause analysis and identifies issues in the supply chain, which could limit on-time customer order fulfillment.

Together, the SAP Fiori planning application view and SAP IBP Excel planning view provide detailed, end-to-end visibility as well as easy-to-use and efficient tools to maximize service levels and minimize waste in the supply chain through cutting-edge planning applications.

The planning view apps you're most likely to use are as follows:

- Demand by Priority (Figure 10.5)
- Projected Stock Analysis (Figure 10.6)
- View Order Confirmation (Figure 10.7)
- Gating Factor (Figure 10.8)

These views are highly useful for efficient planning and review purposes. Information represented in the views can be configured to display relevant data elements such as material, location, customer, quantities, and order number. For ready reference and quick usage, these views have been fortified by graphical representations and colored traffic light signals to represent excess, shortages, confirmed quantities, and unconfirmed quantities.



Rank	Segment ID	Order Type	Order Number	Ite...	Material Number	Location Number	Customer Number	Confirmed Q...
1	GB_SEG_DATE_FIX	Sales Order	721	10	SN1_PHONE_A	DC1	SN-CUST01	100
2	GB_SEGMENT_1	Sales Order	988	10	TF1_PHONE_A	DC1	CUST01	100
3	GB_SEGMENT_1	Sales Order	1012	10	JT61_PHONE_B	DC72	CUST01	100
4	GB_SEGMENT_1	Sales Order	1014	10	JT61_PHONE_B	DC72	CUST01	100
5	GB_SEGMENT_1	Sales Order	1111	10	TF1_PHONE_A	DC1	CUST01	100
6	GB_SEGMENT_1	Sales Order	1124	10	TF1_PHONE_A	DC1	CUST01	100
7	GB_SEGMENT_1	Sales Order	1346	10	TF1_PHONE_A	DC1	CUST01	100
8	GB_SEGMENT_1	Sales Order	1346	20	TF1_PHONE_A	DC1	CUST01	100
9	GB_SEGMENT_1	Sales Order	1736	10	TF1_PHONE_A	DC1	CUST01	100
10	GB_SEGMENT_1	Sales Order	1736	20	TF1_PHONE_A	DC1	CUST01	100
11	GB_SEGMENT_1	Sales Order	1736	30	TF1_PHONE_A	DC1	CUST01	100
12	GB_SEGMENT_1	Sales Order	1736	40	TF1_PHONE_A	DC1	CUST01	100
13	GB_NEW_DATE_FIX2	Sales Order	698	10	TG01_MRP	0001	TG_CUST1	100
14	GB_NEW_DATE_FIX2	Sales Order	865	10	MM1_PHONE_A	DC1	CUST03	100
15	GB_NEW_DATE_FIX2	Sales Order	880	10	MM1_PHONE_A	DC1	CUST02	100
16	GB_NEW_DATE_FIX2	Sales Order	990	10	PAL_PHONE_A	DC1	TG_CUST1	100

Figure 10.5 Demand by Priority App in SAP IBP for Response and Supply

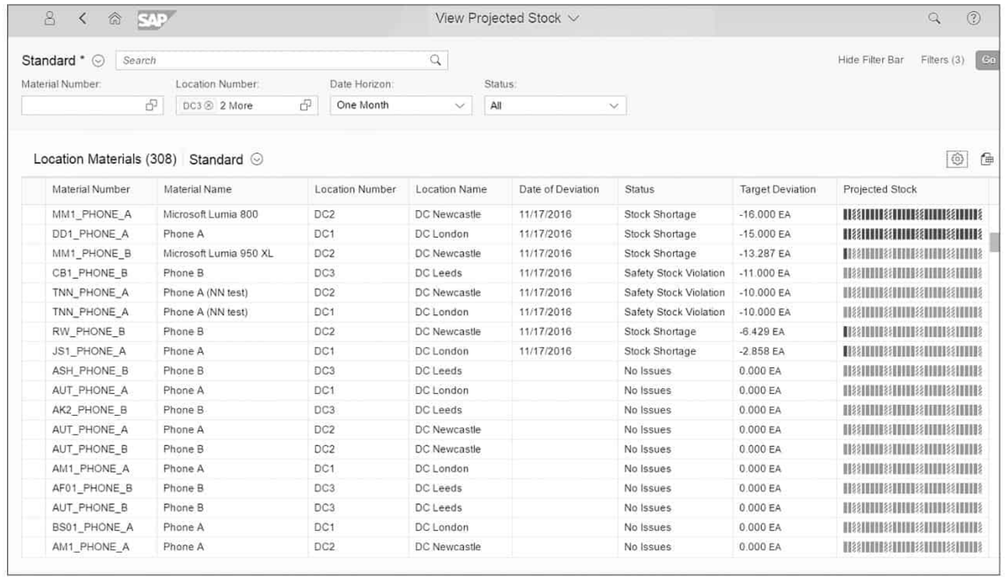


Figure 10.6 Projected Stock Analysis App in SAP IBP for Response and Supply

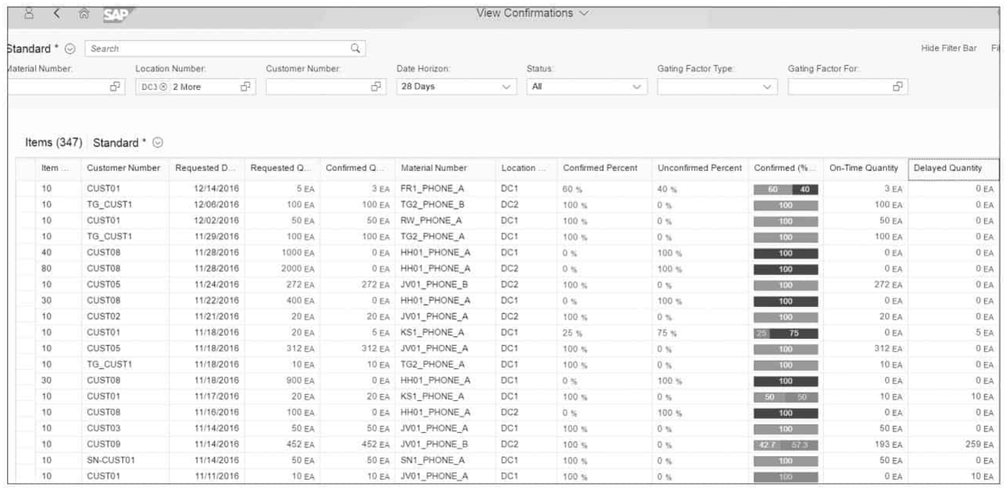


Figure 10.7 View Order Confirmation App in SAP IBP for Response and Supply

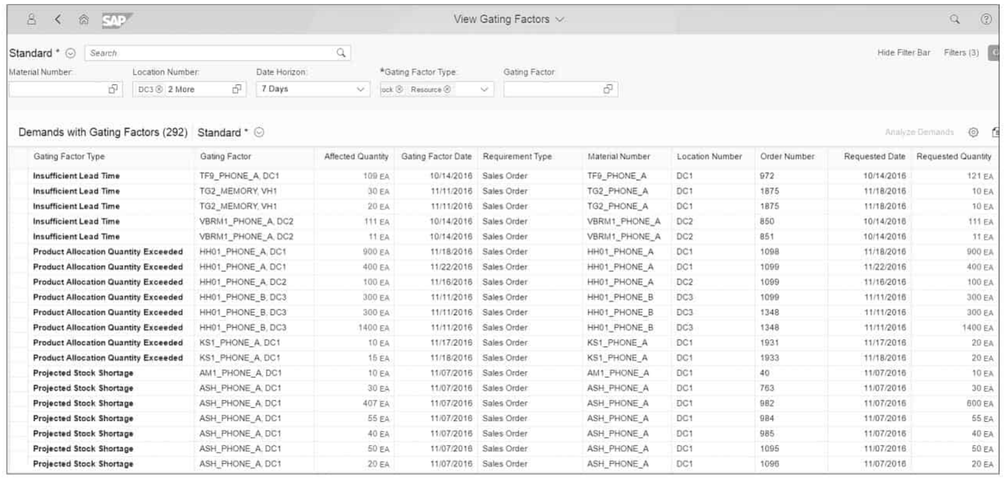


Figure 10.8 Gating Factor App in SAP IBP for Response and Supply

10.6 Order Simulation and Scenario Planning

Simulations in response and supply planning can perform what-if analysis for assessing potential opportunity or risk scenarios without impacting the live data in the system. The functionality in SAP IBP for response and supply allows simulating sales orders, constraints, creating a scenario, collaborating, and choosing the best possible action for supply.

The functionality for using simulations and scenarios can be grouped, as follows:

- Simulation of a sales order
- Scenario creation, analysis, and constraint management
- Collaboration, decision, and data update to the live system

10.6.1 Simulation of Sales Order

With the Simulate Sales Order app in SAP IBP, the system can create an additional sales order and analyze its impact on the current demand and supply situation. A sales order with a material, delivery location, relevant customer, quantity requested,

and priority can be created in the simulated environment. An order added to the system has an impact on the current demand and supply situation; based on this impact, order confirmation for the created sales order as well as for previous sales orders in the system could be affected.

The View Confirmation app shows the confirmation for the created sales order as well as any impact on the existing orders in the system. Product allocation information can also be reviewed and analyzed in the simulated environment. Based on the results displayed by adding the order addition to the simulated environment, along with the planner's expectation, either the simulation can be performed on the live data or further action on a constraint review or a scenario analysis can be carried out to identify the best option. SAP IBP provides a system-generated instruction sheet for the steps taken by the planner in the simulated environment so that the same actions can be performed with live data if required. Detailed system steps for a sales order simulation can be found in Chapter 11, Section 11.6.

10.6.2 Scenario Creation, Analysis, and Constraint Management

A simulation in SAP IBP for response and supply can be saved as a scenario; a scenario with its inherent nature in SAP IBP can be further reviewed, edited, and shared with other users for performing further analysis. Data in the scenario can be reviewed, and based on the supply levers of the organization, it can be edited for impact analysis. For example, for a resource constraint, running an extra shift might be considered, whereas for a supplier constraint, an alternate supply option might be considered. Based on the possibility of potential action to resolve an issue, the action result (e.g., increased machine availability, higher allocation amount, etc.) can be updated in the scenario by updating the supply constraint. A further planning run for the scenario can be performed with an impact analysis in a what-if environment. Figure 10.9 and Figure 10.10 shows an example in which a sales order simulation is being saved as a scenario for further analysis. The data in the scenario can be analyzed through the Manage Version and Scenario app.

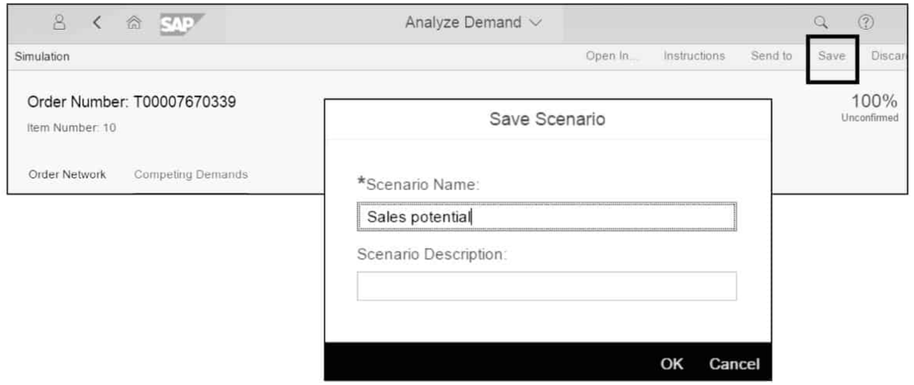


Figure 10.9 Sales Order Simulation to Scenario Creation: Part 1

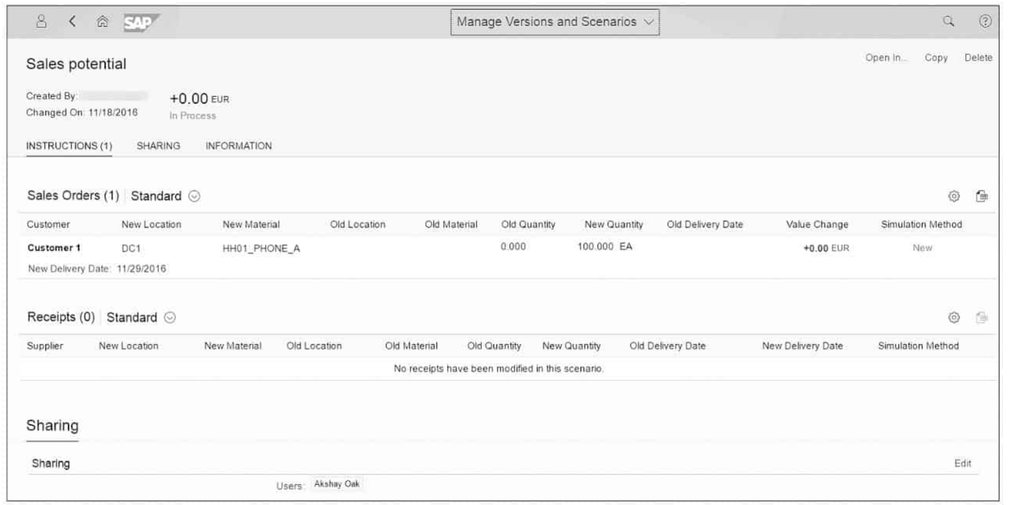


Figure 10.10 Sales Order Simulation to Scenario Creation: Part 2

10.6.3 Collaboration, Decision, and Data Update to the Live System

A planner can share scenario data with team members for collaboration and to determine the best course of action. Figure 10.11 shows how scenario information can be sent to another user in the system for collaboration. The user can review and edit the

information for further analysis. Following the result and impact analysis through order simulation and scenario, a decision can be made to take action against the market opportunity. Based on the result identified and agreed on through the simulation and scenario, an instruction sheet is automatically created by the SAP IBP system that can be downloaded by the user to perform the agreed-upon action in the system. Figure 10.11 shows an example of the instruction sheet as generated by SAP IBP for response and supply. Following the instruction sheet generated by the system, the data can be updated in the live system to make the simulation model part of active data element.

This capability maximizes the profitability from a potential opportunity while considering the impact and agreeing on the best possible option through real-time result analysis and collaboration.

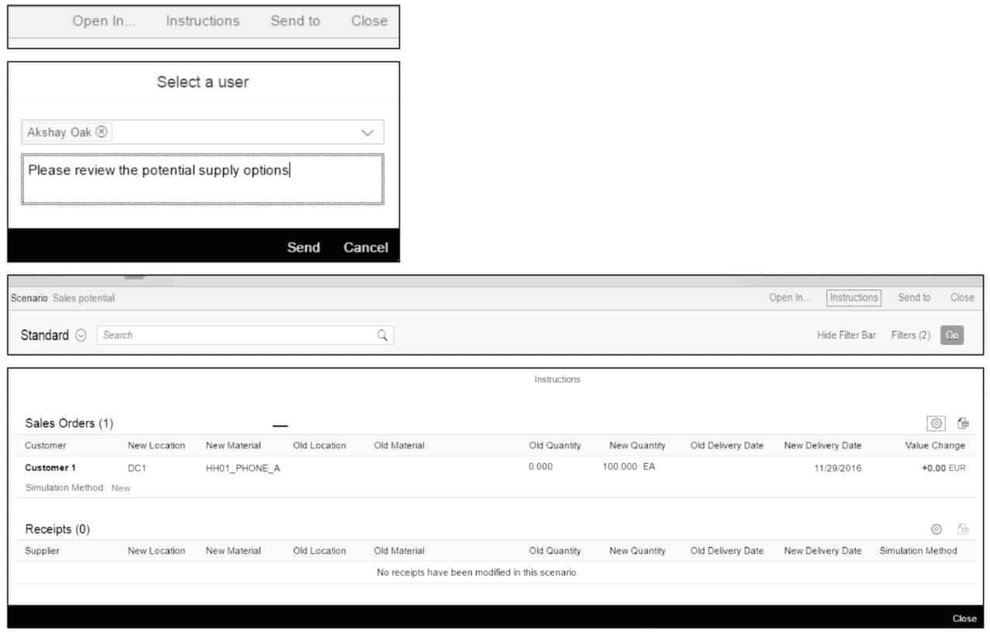


Figure 10.11 Collaboration, Decision-Making, and Instruction Sheet

10.7 What's Ahead for Response and Supply Planning?

The following are some key capabilities on the 2018 development roadmap for SAP IBP for response and supply:

- **Operational supply planning**
  The following operational supply planning functionality is planned for future SAP IBP releases:
  - Optimization
    Additional optimization functionality will be added, including the following features:
    - The ability to optimize the operational supply plan based on costs at the order level to improve profitability
    - Detailed pegging of supply and resource usage to enable analysis and resolution of constraints at the order level
  - Planning in multiple planning areas in a single SAP IBP instance to support segmentation of the supply chain by different dimensions such as business unit
  - The ability to model and plan using multiple modes of transport such as air, ship or rail
- **Tactical supply planning**
  The following tactical supply planning functionality is planned for future SAP IBP releases:
  - Forecast consumption logic that decrements the forecast by actual demand to allow a more accurate supply plan
  - A shelf-life planning heuristic that will take into account product expiration date when planning replenishment
  - Master data simulation in user-defined scenarios will allow end-users to add or change master data such as capacity, customer or product attributes to simulate the impact on the supply plan

Note

For additional information on the SAP IBP roadmap, see *https://bit.ly/2pQOtGc*.

10.8 Summary

Supply to fulfill customer demand is created through production, purchase, or stock transfer activities by the organization. Supply planning considers demand, current inventory, and sourcing rules. An agile supply network has the flexibility to respond in the short term against market realities through response planning capabilities.

Supply planning algorithms, unconstrained heuristic planning, finite optimization, and rule-based supply planning are used in SAP IBP for response and supply to generate supply plans. The response planning algorithm automatically performs order prioritization and product allocation, following the defined rule for creating the most suitable supply plan for the organization. Simulation and collaboration capabilities of SAP IBP are applied for further review and finalization of the supply plan.

Now that we have a good understanding of supply planning processes and the capabilities of SAP IBP for response and supply, the next chapter will discuss how to configure the system to enable these capabilities and processes in detail.

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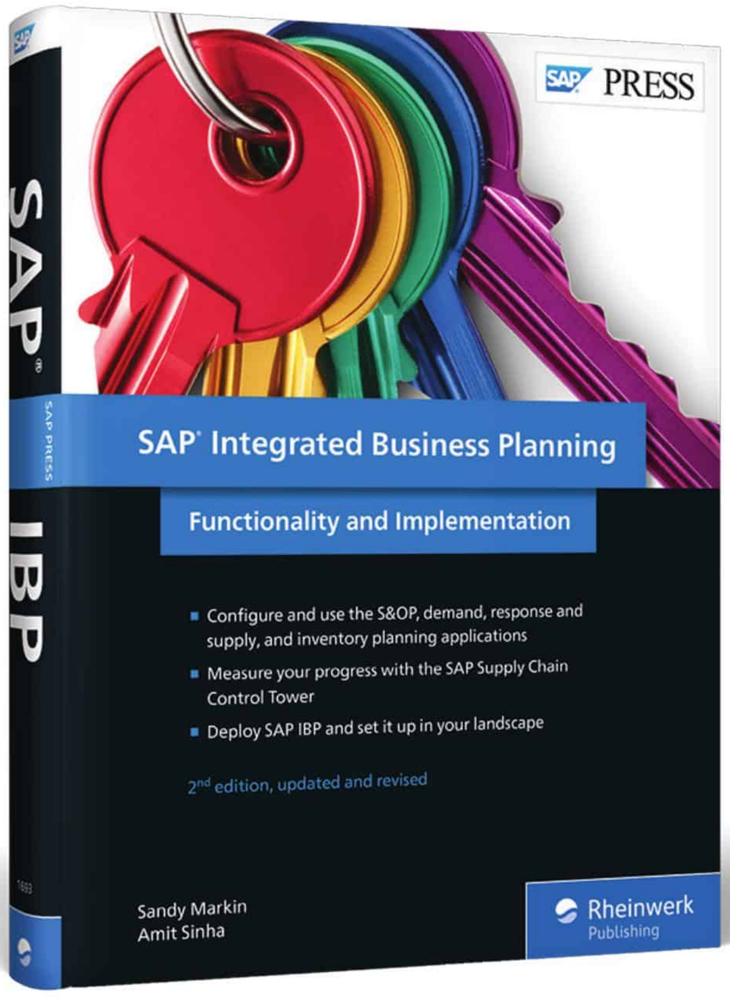
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